

Ayman Patel

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Professional Summary

Backend-focused software engineer with 6 years at Mastercard building production systems across Java, Spring, microservices, event-driven architecture and React. Delivered APIs, resilient services, CI/CD and production observability in payments. Built Golang microservices with gRPC, OpenAPI and Envoy, and deployed distributed workloads with Docker and Kubernetes.

Core Technologies

Backend: Go, Java, Spring, Spring Data JPA, Hibernate, REST APIs, microservices, event sourcing, CQRS | **Data and messaging:** Relational and NoSQL databases, PostgreSQL, Oracle, MongoDB, Redis, Kafka, RabbitMQ | **Platform:** AWS, Docker, Kubernetes, Jenkins, Gradle, OpenTelemetry, Grafana, Splunk, Dynatrace

Work Experience

Nebius Academy

AI Engineering Fellow

London, UK

Mar 2026 – May 2026

- Deployed distributed data-parallel training on Kubernetes across two Nvidia H100 nodes, using SkyPilot and `torchrun` to coordinate reproducible workloads.
- Containerised training pipelines with Docker and Nebius Container Registry; moved tunable configuration into YAML to support repeatable experiments without code changes.
- Diagnosed distributed failures by tracing NCCL network initialisation and node connectivity, separating infrastructure issues from model configuration errors.
- Cut LLM generation time from 0.943s to 0.141s for 128 tokens by implementing KV caching and preallocated sequence buffers, a 6.69x speedup.

Mastercard

Senior Software Engineer

Vadodara, India

Mar 2024 – Aug 2025

- Developed backend APIs with Spring, Spring Data JPA and Hibernate, using an API Gateway, Kafka and event-sourcing patterns for asynchronous, auditable flows.
- Implemented Resilience4j circuit breakers and retries across 5+ microservices, reducing cascading failures and improving system availability by 15%.
- Used PostgreSQL JSONB for dynamic forms, allowing business teams to add fields without database migrations; added Redis caching to reduce repeated API calls.
- Provided on-call support for production issues and used Splunk, Grafana and Dynatrace to diagnose and resolve incidents, cutting mean time to resolution from days to hours.
- Owned the architecture and rollout of a Webpack Module Federation microfrontend platform, enabling 10+ product teams to deploy independently and cutting release coordination from days to hours.
- Led the host-shell migration from legacy Redux to Redux Toolkit and RTK Query; used Vite, Vitest and esbuild across the frontend toolchain.
- Used Playwright for end-to-end testing and supported SDETs in setting up CI infrastructure for automated test execution.
- Led cross-country delivery across engineering teams, translating product requirements into technical specifications, user stories and acceptance criteria.
- Worked in Agile Scrum through sprint planning, daily stand-ups, retrospectives, backlog refinement and code reviews.
- Led Scrum ceremonies when the Product Owner or Scrum Master was unavailable, keeping the team aligned on its sprint goal.
- Coached 10+ engineering teams, documented architecture and delivery practices, and explained design decisions to engineers, product stakeholders and non-specialists.

Software Engineer II

Mar 2022 – Feb 2024

- Built greenfield backend services with Spring, Oracle and Hibernate.
- Owned integration of payments with subscription support for Mastercard Trends, which drove revenue for the programme.
- Owned TDD for backend development using JUnit, Testcontainers and Mockito, and enforced it across multiple teams through training.
- Built greenfield React and TypeScript frontends with legacy Redux state management.
- Ejected Create React App to implement a custom Webpack configuration and maintained the resulting build pipeline.
- Followed the testing pyramid using Jest and React Testing Library for unit tests, with Cypress for automated end-to-end coverage.
- Built Gradle pipelines for dependency management and artifact generation while maintaining performance, code quality and release reliability.
- Achieved 95% test coverage and cut end-to-end CI runtime from 8 hours to 3 through parallel execution, test splitting, selective retries and caching.
- Owned Mastercard's accessibility initiative, lifting WCAG compliance by 40% and contributing 5 fixes and additions to the internal design system.

Work Experience (continued)

Mastercard

Vadodara, India

Software Engineer I

Jul 2019 – Feb 2022

- Designed REST APIs to automate business workflows with Camunda and modelled complex rules using BPMN, DMN and CMMN.
- Developed and maintained an HSM integration for MDES tokenisation, securing card-data transmission across payment flows.
- Reviewed penetration-test reports, prioritised findings by risk and delivered remediation for enterprise security requirements.
- Designed a Jenkins CI/CD pipeline for an Angular.js application using Grunt.js.

Software Engineering Intern

Jan 2019 – Jun 2019

- Built an asynchronous image-serving backend for MDES CardArt with Spring WebFlux and Reactive Streams.
- Compressed images with ImageMagick to reduce payload size and improve response times; used TDD to support reliable changes.

Personal Projects

Go Microservices

Go, gRPC, OpenAPI, Envoy

- Built an end-to-end Go microservices stack with Gorilla Mux request validation and OpenAPI documentation.
- Connected services with gRPC and configured Envoy for Layer 4 and Layer 7 load balancing.

Repository Summarizer

FastAPI, LLMs, Tree-sitter

- Built a FastAPI service that uses abstract syntax trees and Tree-sitter to extract repository context within a token limit and generate readable GitHub repository summaries.

Java CLI

Java, GraalVM, Picocli

- Built a Java CLI with Picocli and compiled it to a cross-platform native binary with GraalVM for sub-zero startup time.

Education

Queen Mary University of London

London, UK

MSc Advanced Computer Science

Sep 2025 – Sep 2026

Sardar Vallabhbhai Institute of Technology

Vadodara, India

Bachelor of Computer Science, GPA: 7.9/10

Jul 2015 – May 2019

Technical Skills

Backend and architecture: Go, Java, Spring, Spring Data JPA, Hibernate, Python, FastAPI, REST APIs, gRPC, microservices, API Gateway, event sourcing, CQRS, concurrent and reactive programming

Databases and messaging: PostgreSQL, Oracle, MongoDB, Redis, Kafka, RabbitMQ, Debezium, Change Data Capture

Cloud and delivery: AWS (S3, Lambda, EC2), Docker, Kubernetes, Terraform, PCF, Jenkins, Gradle, SkyPilot

Quality and operations: JUnit, Mockito, Testcontainers, Playwright, Cypress, k6, Grafana, Prometheus, Splunk, Dynatrace, OpenTelemetry

Writing

- Published 40+ technical articles in [my blog](#), covering AI/LLM systems, observability and distributed data infrastructure.
- Wrote accessibility and microfrontend whitepapers that earned CTO recognition, became internal references and reached 200 engineers.
- **Undergraduate seminar - Event-driven architecture using Kafka:** [Report link](#). Delivered a seminar covering Kafka architecture, AMQP, Kafka Streams and Kafka Connect.

Hackathons

{Tech: Europe} Hackathon, London

2026

- Won the Most Secure Build Award from Aikido Security with zero security vulnerabilities in the codebase.
- Built a platform that turned academic papers into investor-ready intelligence and dynamic presentations, using Tavily for research and Attio for agentic CRM.

Encode Hackathon, London

3rd Place, 2025

- Built **Phantom Fraud**, an AI-powered financial risk engine for the Solana blockchain.
- Used Solana CLI, OpenRouter and Helius APIs to parse on-chain transactions and detect wash-trading patterns.

AI Hackathon

2nd Place, 2024

- Developed a RAG-based internal documentation search system using LangChain and Chroma.
- Chunked and embedded internal documents, then returned grounded answers with source references.

Internal Hackathon

2nd Place, 2023

- Migrated a React application to Next.js, improving load performance and enabling server-side rendering and static generation.